Complete permutation searches and the six-point row

Human Compression project

This note proves the complete search reduction for two experimental rows of Tables 1 and 3 of Leemans's survey, and independently settles its six-point symmetric row. The larger numerical conclusions require every listed branch to finish and its complete output to be checked. A bounded or interrupted computation gives no upper bound on the number of markings.

We use the previously proved enumeration theorem for actual permutation markings. Its proof concerns a group G normalized by a permutation group A, rather than alternating groups alone. In the cases considered here, $G = \mathrm{Sym}(n)$ or $\mathrm{Alt}(n)$ and $A = \mathrm{Sym}(n)$. For $n \neq 6$ this action induces the full automorphism group of G. Thus point conjugacy is exactly the isomorphism relation on ordered markings in degrees fourteen and fifteen. The exceptional six-point automorphism is handled explicitly below.

1 The inherited recursion

Let I be the set of nonidentity involutions in G, sorted by the fixed permutation ordering. A node carries a prefix $t = (t_0, \ldots, t_{m-1})$, its actual generated subgroup H, its pointwise stabilizer $S = C_A(t_0) \cap \cdots \cap C_A(t_{m-1})$, and the pool of involutions commuting with $t_0, \ldots, t_{m-2}$. For each S-orbit of candidates outside H, choose its least element. Append that element exactly when the resulting tuple has the intersection property. The new pool is obtained by intersecting the old pool with $C_G(t_{m-1})$. Thus the newly appended mark need not commute with its immediate predecessor, but every later mark must do so.

The intersection test is the acquired recursive criterion: the two end omissions must themselves be string C-markings, and their actual intersection must be the subgroup generated by the common middle marks. The left omission is already certified at the preceding node; the test recursively checks the right one and computes the actual intersection. No quotient is used. A new generator lying in H is forbidden by the intersection property, so its rejection loses no valid extension.

For completeness, the induction on prefixes is short. Every extension can first have its existing prefix conjugated to the representative already retained. The remaining possible conjugacies fixing that prefix are exactly S, so precisely one representative of each extension orbit is visited. Conversely, two prefixes retained in different such orbits cannot be conjugate with the preceding prefix fixed. Induction gives both completeness and absence of repetitions.

The candidate set is invariant under S: every element of S fixes all selected marks, hence H, and preserves every commutation condition imposed by the pool. Thus enumerating the orbits on this whole domain is equivalent to closing the same set of seeds under the action. The replacement of GAP's `Orbits` by `OrbitsDomain` changes only that implementation choice. All six completed small branches have identical full literal outputs under the two routines.

The next-pool evaluation has a further exact implementation simplification. For a sorted pool $P \subseteq I$,

$$P \cap \{x \in I : [x, t] = 1\} = \{x \in P : [x, t] = 1\}.$$

Filtering P directly by this predicate therefore gives exactly the same ordered list as forming the complete ambient commuting set and intersecting it with P. No representative, subgroup

or descendant changes. All nine completed comparison branches retain identical full literal output.

The transitivity rejection is also literal. Every further generator lies in the current pool, which can only shrink. If the subgroup generated by H and that entire pool is intransitive, no descendant generates the transitive target G. A node already generating G has no independent extension. Both rejections therefore preserve completeness. Transitive proper subgroups are retained as well, but only records with actual group order $|G|$ enter the present counts; as subgroups of G, those records then generate G itself.

2 Connected prefixes for the fixed targets

Suppose adjacent marks t_j, t_{j+1} commute. The string relations then make every mark in $\{t_0, \ldots, t_j\}$ commute with every mark in the remaining interval. The two generated subgroups are nontrivial and have trivial intersection by the intersection property. They therefore give a nontrivial direct product decomposition of the whole group.

Neither target group admits such a decomposition for $n \geq 5$. For A_n this follows from its standard simplicity. For S_n, any nontrivial normal subgroup N contains A_n: otherwise simplicity gives $N \cap A_n = 1$, so $[N, A_n] = 1$. But $C_{S_n}(A_n) = 1$, since a permutation centralizing every three-cycle preserves every three-element support, and hence fixes each point. Thus two nontrivial direct factors would both contain A_n, contradicting their trivial intersection. Consequently every adjacent pair in a target marking must fail to commute. The search may reject a prefix at its first commuting adjacent pair; later marks cannot restore a connected diagram across that cut.

This filter removes complete stabilizer orbits, because the stabilizer fixes the preceding mark. All remaining minimum representatives are unchanged. The new implementation enumerates the connected part of the old transitive subgroup list; it is not a complete list of disconnected proper subgroups. Complete small lists are compared after precisely this filter, and all target counts must remain identical. The direct-factor argument, rather than the small controls, proves the uniform target completeness of the filter.

3 Independent initial branches

Lemma 1. *For $G = \mathrm{Sym}(n)$, the nonidentity involution classes under $A = \mathrm{Sym}(n)$ are indexed by $1 \leq e \leq \lfloor n/2 \rfloor$, where e is the number of transpositions. For $G = \mathrm{Alt}(n)$, retain exactly the even values of e. Let u_e be the first member of I having e transpositions. The entire branch beginning with u_e is obtained by starting the inherited recursion at*

$$((u_e), \langle u_e \rangle, C_A(u_e), I).$$

Proof. Two involutions are conjugate in the symmetric group precisely when their cycle types agree, and the cycle type is $2^e 1^{n-2e}$. Such an involution is even precisely when e is even. The sorted first member is the least element of this orbit, exactly the representative chosen at the empty prefix. The inherited recursion imposes no commutation restriction on the second mark, so the child pool is still all of I. Its subgroup and stabilizer are exactly the displayed groups. Every subsequent operation is unchanged. Different values of e cannot produce conjugate ordered markings because their first entries have different cycle types. □

The direct entry removes only the computation of all initial orbits. It does not change candidate selection or the tests below that entry. The original whole search, its version selecting one initial orbit after forming them all, and the direct entry have been compared on S_5, S_7 and A_5. Their complete literal lists agree after taking the appropriate union. The known counts

provide additional controls. These checks support the implementation; the lemma supplies the all-degree justification.

Each initial branch has a separate immutable driver and a bounded execution receipt. Completed branches may be reused without rerunning them. A cap records incompleteness. Numerical admission requires one complete receipt for every admissible value of e, with the same group, rank range and engine edition. A separately documented equivalent entry edition may be combined only through the lemma and exact provenance; an unfinished predecessor is never counted as a second branch.

4 Rank ranges and reversal

For S_{14}, the only unacquired ranks are $3 \le r \le 8$. The accepted stability theorem supplies ranks 9 through 13, since with $k = 14 - r$ its condition $14 \ge 2k + 3$ holds. The accepted independent generating-length bound excludes ranks at least 14. For A_{15}, the accepted alternating maximal-rank theorem gives $r \le 7$. Hence the fixed searches require seven initial branches for S_{14} through rank eight and three initial branches for A_{15} through rank seven.

Let c_r be the resulting number of ordered classes and s_r the number fixed by reversal. For each literal representative the latter condition is tested by asking whether a permutation in $\mathrm{Sym}(n)$ conjugates its entire ordered tuple to its reverse. The number of classes with dual markings identified is

$$(c_r + s_r)/2.$$

This follows from the orbits of the reversal involution on the complete set of ordered classes. Reversal can exchange distinct initial branches; therefore this quotient is formed only after all branches are combined. The two large target groups have no exceptional outer automorphism affecting this calculation.

5 The complete six-point calculation

Proposition 2. *For S_6, the numbers of string C-markings of ranks $3, 4, 5$, up to group isomorphism and reversal, are respectively $2, 4, 1$.*

Proof. All three initial involution classes are exhausted through rank five. The recursion gives $6, 14, 2$ ordered classes under point conjugacy. The literal representatives are included with every subgroup and output receipt. In addition to the recursive enumeration test, a separate verifier checks every pair of subsets of each representative's index set, requiring that the intersection of the two actual generated subgroups equals the subgroup on the common indices. It also checks generation of S_6 and all involution and string relations. There are 6016 such subset pairs.

Point conjugacy is not the full equivalence relation in degree six. We use the standard fact that $\mathrm{Aut}(S_6)$ has order 1440, with its inner subgroup of order 720. The certificate constructs the full automorphism group and an actual bijection ϕ taking a transposition to three disjoint transpositions. It verifies that the inner automorphisms and ϕ generate the full group. For every retained tuple, conjugacy under S_6 locates the unique retained class of its ϕ-image and of its reverse. These give two literal permutations on each complete class list. Their orbits, not a point-conjugacy formula borrowed from another degree, give the required numbers:

rank	3	4	5
point-conjugacy classes	6	14	2
full automorphism classes	3	7	1
full automorphisms and reversal	2	4	1

The exact two permutations are retained in the certificate. Inner conjugacies act trivially on the class list, so the group generated by these two permutations is precisely the required equivalence relation. The accepted independent generating-length bound gives rank at most five for S_6, which completes this row. □

The number four in the middle column is the previously unacquired rank-four cell of the survey. The other two values are fresh checks of acquired cells. The input does not use these expected counts: complete lists are generated first, and source comparison follows. The six-point controls also test the connected-prefix change on proper transitive subgroups: exactly seventeen disconnected markings are removed across the three initial classes, while all twenty-two target markings remain unchanged.

The search and its proof are reused from the acquired finite-census method. The direct initial entry, invariant-domain evaluation, direct-factor pruning, the pool evaluation and exact survey-cell interfaces are the additional work described here. The source's numerical answers do not direct the search and are compared only with completed results. This component supplies no claim about an open conjecture, no new measurement, and no whole-source admission.